
TECHNICAL PROPOSAL FOR DISCUSSION

Fabric Vitals: a vendor-neutral health score for AI data-center fabrics, measured against a declared design

A proposed health score for AI data-center back-end scale-out fabrics. It measures how well the fabric is keeping the promises on its Fabric Spec Sheet, under a common Rulebook.

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Fabric Vitals v0.2 · a technical proposal for discussion. Not a standard. Rulebook values not yet set.

Specification and open questions: github.com/fabric-vitals/spec · **Contact:** hello@datacenternetwork.ai

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Summary

Fabric Vitals is a proposed health score for AI data-center back-end scale-out fabrics. It measures how well a fabric is keeping the promises on its Fabric Spec Sheet, under a common Rulebook, and produces one headline number, the **Vitals Score**, with the reasons underneath. Operators already use health dashboards, intent/assurance systems and many separate signals; the narrower gap this proposal addresses is a common vendor-neutral operational conformance method built around that shared measurement contract and pre-declared promise.

The method works like the rating of a car's range. Every car is measured by the same official test procedure, but each car has its own rated range. In Fabric Vitals, a **Rulebook** says how fabrics are measured and judged, and a **Fabric Spec Sheet**, declared before measurement, says what this fabric promises. The score is therefore a conformance reading rather than an absolute quality or risk ranking. The credit-score analogy is about presentation only: one number at a glance, with reasons underneath. Four design choices shape the proposal:

- **Seven vitals** are measured from evidence that comes from the network itself, not from the jobs running over it.
- **The v0.2 headline is the weakest required vital.** A short list of named critical conditions can force it lower still. This prevents compensation in the headline by construction, but it is a falsifiable proposal: E4 and E5 compare the current minimum, a calibrated minimum, an arithmetic-average comparator and other serious alternatives against measured network-attributable impact, explainability and robustness.
- **Unmeasured parts of the fabric count as unknown, not healthy.** Switching some of it off, or leaving part of the fabric unmonitored, can only lower the score, never raise it.
- **Every rule and every promise is fixed and published before measurement**, so nothing can be tuned afterwards to make a result look better.

This paper defines the method. It does not yet set the Rulebook's numerical values, such as the exact thresholds: those are to be fixed through an open validation programme, a public series of experiments in which anyone can take part. Readers who want to critique the method, report which measurements their equipment can export, or offer a fabric for testing will find how in the closing section, "How to take part".

1. The problem

The people who operate AI fabrics already combine many signals: link state, drops, ECN marks (the flag a switch sets to warn senders that a queue is filling), PFC pauses (the signal a switch sends to stop its neighbour before a queue overflows), buffer occupancy, retransmissions, latency tails, forward-error-correction statistics, optics readings, routing state, and synthetic collective tests (test traffic that imitates the way GPUs exchange data during a job). Existing health and assurance systems are useful, but there is no widely adopted common vendor-neutral AI-fabric operational conformance method built around a shared measurement contract and pre-declared fabric promise. Current approaches include:

- **pass/fail gates**, such as node and switch health checks and acceptance tests [1, 2, 3];
- **each operator's own thresholds**, for example Meta's PFC watchdog, which acts when a pause lasts longer than 200 ms, and its buffer alarms, which fire when a buffer is more than 80% full [4];
- **deviation from a baseline or from peers** [5, 6, 7];
- **vendor dashboard scores** whose inputs and weights are only partly documented [8, 9, 10].

These answers share three weaknesses.

They are not comparable. Each operator's thresholds and baselines are its own, so a score from one operator cannot be set beside a score from another. A baseline-relative "healthy" means "unchanged", not "built and running

as designed". A fabric that has been degraded since day one becomes its own normal.

Averages can mask localized failures. Several vendor scores average their inputs [9, 11]. In the simple masking example, four areas at 100 and one at 10 produce an arithmetic mean of 82, the same mean as five areas at 82. The OECD/JRC handbook on composite indicators describes this compensation effect and illustrates it with its (21, 1, 1, 1) versus (6, 6, 6, 6) example [12]. This example motivates a non-compensatory candidate; it does **not** by itself prove that arithmetic averaging is the wrong overall aggregator. E5 compares both approaches, and others, against measured operational impact and robustness criteria.

They do not say what the network was supposed to be. Losing one uplink is routine in a fabric designed to tolerate two link failures, and serious in a single-homed fabric, where each NIC is attached to only one switch. In one 256-GPU testbed, training "cannot recover" when the repair of a single-homed link takes more than two minutes, while a dual-homed design saw 6.25% degradation [13]. In a multi-plane design (several independent parallel networks) with packet spraying (each NIC scatters its packets across all of them), the job continues through a lost NIC-to-switch link [14]. The same physical event means different things depending on the design and on the transport.

2. Why AI fabrics need their own health measure

A characteristic of AI fabrics justifies a dedicated measure only if it changes what is measured, how it is normalized or how it is aggregated. Six do.

1. **Synchronized collectives are gated by the slowest transfer** [13, 14, 15]. A collective is an exchange in which every GPU in a job sends data to the others and waits until all of it has arrived. A single bad link can slow a whole training step, so the aggregation must be tail-dominated and non-compensatory.
 2. **Congestion behaviour depends on the transport.**
 - AI fabrics carry RDMA traffic (remote direct memory access, a technique that lets one server write straight into another server's memory across the network without the receiving processor handling each packet). Run "lossless", with switches pausing rather than dropping, RDMA over Ethernet brings PFC head-of-line blocking, pause storms and deadlock [16, 17, 18].
 - The available signals depend on the congestion-control mode. Without DCQCN, the congestion-control scheme for RDMA over Ethernet that reacts to ECN marks, Meta monitors NIC pause duration instead of congestion-notification counts [4].
 - The same loss rate means different health under different transports. In a two-server test, a deliberately injected loss rate of 0.4% gave zero goodput under go-back-0 retransmission, a recovery scheme that resends a message from its first packet after any loss [16]. A multipath spraying transport with selective retransmission rides out tier-to-tier link flaps [14].
 3. **Low-entropy elephant flows and bursts.**
 - ECMP hash collisions limited RDMA to 60% utilization in one Microsoft experiment across two podsets of a new data center [16].
 - Bursts saturate 400G NICs for seconds [13].
 - At four-minute polling, utilization barely predicted drops ($r = 0.098$, in a general data-center study) [19].
- Capacity evidence must therefore be burst-aware, and load-imbalance metrics apply only to flow-hash fabrics.
4. **Gray failures are hidden by redundancy.** A gray failure is a device that reports itself healthy but is not doing its job.
 - In one production job on about 75,000 GPUs, a switch that stayed "up" but stopped forwarding dropped about

580,000 packets without device state showing it [14].

- Silent drops and black holes appear only to active probing [5].
 - Standalone stress tests catch degradation that monitoring of running workloads misses [1].
5. **Rail, multi-plane and rail-only topologies.** In a rail design, the first GPU in every server connects to one switch, the second GPU in every server to another, and so on; each such group is a rail.
- Rail-only fabrics are *meant* to lack any-to-any paths [20].
 - In one published design comparison, losing a single tier-to-tier link removes 3% of a node's capacity in an 800G plane (a plane built from faster links) but 0.4% in a 100G plane (built from slower links); design arithmetic, not measurement [14].
 - A vendor reference design already prescribes dual planes at 256 GPUs [21].

Health must therefore be normalized to capacity share and to the declared design.

6. **Training and inference differ.** Collective tail and throughput dominate training [13, 15]. In disaggregated inference, where reading the request and producing the answer run on different servers, what matters is time-to-first-token, time-per-output-token and the transfer of the KV cache, the working memory built while reading the request [22, 23].

Everything else is shared with generic network health. That is where existing standards are reused rather than replaced. See Section 7.

3. Design principles

Principle 1: measured against a declared reference. A health score is a confirmation of conformance to an agreed standard. An AI fabric is designed to a level of resilience and recovery, and the score confirms whether it meets that level. Every metric and threshold either comes from an agreed standard or best practice, or is explicitly defined, with documented reasoning, before anything is measured. How fabrics are judged is fixed in the Rulebook; what each fabric promises is fixed on its spec sheet. No value is tuned after the fact.

Principle 2: simple at the top, deep underneath. One number, always shown with its spec sheet and class, its range, the reason it has its value, and the full set of vitals.

Principle 3: no false health. One catastrophic condition cannot be concealed by several excellent results elsewhere.

Principle 4: unknown is not healthy. Unmeasured capacity lowers the *confirmed* score, the part the evidence actually supports. It never raises it.

Principle 5: no false precision. Scores are whole numbers only. Colour bands are a display rule, published only once calibrated. Every parameter without evidence is marked "value not set".

Principle 6: designed to be vendor-neutral. The method describes behaviour, not products. Where common telemetry models do not exist, the dependency and required vendor/platform adapter are stated. Complete cross-vendor measurement equivalence is a validation target, not an assumed property.

Principle 7: deterministic. The same telemetry, spec sheet and Rulebook always produce the same score. Language models may explain a score but never compute it.

4. How Fabric Vitals works

4.1 Two inputs: the Rulebook and the Fabric Spec Sheet

Every car's range is measured by the same official test procedure, but each car has its own rated range, printed on its spec sheet. Nobody expects a small city car and a long-distance saloon to have the same range; everybody expects both to be measured the same way. Fabric Vitals makes the same separation.

The Rulebook (identical for every fabric). The Rulebook (called the reference profile in v0.1) says how to measure and how to judge: the metric definitions and measurement procedures; the coverage floor, the minimum share of the fabric that must be measured before a score can be given; the blast-radius threshold, the share of capacity a fault must affect before it counts as critical; the grace period, how long a late measurement is still trusted; the critical-condition definitions and cap; the rules for UNSCORABLE; the temporal windows; the score band edges; and the versioning rules. It is published and frozen before measurement, and it is the same for every fabric.

The Fabric Spec Sheet (declared per fabric, before measurement). The owner fills in a standard template with 23 fields:

- **purpose and workload profile:** training, inference or mixed;
- **scoring scope:** the unit within which jobs run, a pod or scalable unit (a block of servers and switches built and operated as one), which is scored as one. See Section 4.10.
- **build:** topology, host attachment, uplinks and subscription ratio, planes, transport (flow hashing or spraying), the lossless-transport configuration, and physical diversity;
- **resilience promise:** the failures the fabric rides through without capacity loss, and its recovery objective, including routing convergence time;
- **availability target:** the share of time the Vitals Score is expected to stay in the healthy band. See Section 4.8.
- **performance envelope:** bandwidth per accelerator, collective bandwidth reference, tail-delay budget, loss and pause budgets, and physical-layer budgets;
- **required tests:** whether synthetic collectives are required, and the telemetry the fabric exports;
- **declared class** (a preset or custom) and the spec-sheet version. See Section 4.2.

Each vital checks spec-sheet fields.

Vital	Spec-sheet fields it checks
D1 Reachability	Scope and inventory; topology; host attachment
D2 Resilience	Host attachment; uplinks and subscription; planes; failures ridden through; recovery objective, including convergence time
D3 Congestion and loss	Transport; lossless-transport configuration; loss budget; pause budget
D4 Delay and tail	Workload profile; tail-delay budget
D5 Physical link integrity	Physical-layer budgets
D6 Effective capacity	Uplinks and subscription; transport (hashing or spraying); bandwidth per accelerator
D7 AI communication	Collective bandwidth reference; required tests

The register split. The register is the full list of parameters the method uses. Every parameter in it is labelled Rulebook or Spec sheet. Parameters that are really per-fabric promises (the pause budget, NIC retransmission budget, early-warning margin for bit errors, residual capacity after failures, recovery objectives, the class values themselves, and the physical-layer budgets as a standard floor) move to the spec sheet, with the status VALUE DECLARED BY OWNER. The tail-delay budget and the collective bandwidth reference keep their measurement method in the Rulebook and take their value from the spec sheet. Everything that decides *how we judge* (coverage

floor, grace period, blast-radius threshold, the coupling rule between them, the cap, event definitions, band edges) stays in the Rulebook. Of 40 parameters, 31 are Rulebook, 7 spec sheet and 2 both.

The spec sheet can promise less, but it cannot hide what it promises. The declaration is public in the score record, and comparisons are allowed only between fabrics on the same class or on identical spec sheets. See Section 4.9.

4.2 Class presets and the conformance check

Class presets are proposed ready-made Spec Sheets. Fabric Vitals v0.2 publishes three candidate reference profiles to give engineers concrete starting points for critique and testing. Each is a complete Spec Sheet with **structural predicates** (the build features a fabric of that class must have) and **promise fields** (what it promises). The classes are design patterns ordered by redundancy, not quality tiers.

Class	Name	What it is
R1	Rail-optimized, single-homed	A documented training design pattern appearing in multiple current reference designs and operator reports. Each accelerator NIC attaches to one leaf switch on its rail; recovery from a leaf failure is by checkpoint restart.
R2	Dual-homed	Each NIC attaches to two switches, so the fabric rides through a leaf failure, or a limited number of uplink failures per rail, without losing capacity.
R3	Multi-plane, sprayed, tested	Several independent planes, packet spraying across them, and continuous synthetic collective tests.

The cited evidence documents the selected patterns and their component design choices. In one vendor's validated design, "rail Nth connects all GPUs in position Nth on all the servers, to leaf node Nth" [24], and another vendor's reference compute fabric "is rail-optimized" [25]. Alibaba describes single-ToR attachment as "widely used in the majority of current cloud providers" [13]. Alibaba's own production design "connects two ports of each NIC to different ToRs in an active-active way", so that "if one ToR (or a port) is down, the other can still work" [13]. Multi-plane designs with packet spraying ride through tier-to-tier link failures [14].

An owner may instead declare a custom spec sheet; its score is then comparable only with fabrics on identical spec sheets. The preset's name fixes its pattern. All numeric preset values are proposed and not set, except the physical-layer budgets, where the IEEE budget applies. See Section 4.3.

Why failure consequence, not path counts. Facility standards count distribution paths [26, 27]. Rail-optimized AI designs, however, attach each NIC to a single rail switch by design, and one open reference design states that "no redundancy is assumed or required" [28]. AI resilience comes instead from planes and spraying [14], from dual top-of-rack attachment [13] or designed spare capacity. In one AI zone, Meta chose an under-subscription ratio "to allow for buffer for up to two link failures" [4]. Presets are therefore stated in terms of *failure consequence*: what happens to the job when a given failure occurs.

The borrowed ideas and the proposal. Four ideas come from data-center facility practice:

- the owner declares the required class before design, based on the business impact of an outage [29];
- rating by the weakest subsystem: "The site's Tier rating is the lowest of the individual subsystem ratings" [30];
- no fractional ratings [30];
- outcome-based confirmation tests [30].

The preset scheme itself is the author's proposal, derived from public sources, and offered for critique.

The build is checked against the fabric as built. The **verified class** is the most redundant preset whose structural predicates all hold. It is checked three ways: by static comparison of the declared build with the discovered topology and running configuration; by a residual-capacity computation under the declared tolerated failures; and by periodic

controlled-failure tests measured with synthetic collectives, which also measure recovery and convergence time against the recovery objective. The **conformance status** is CONFORMANT, NON-CONFORMANT (with the failed predicates listed) or NOT ASSESSED.

The score is computed against the declared spec sheet, and over-claiming is visible. Redundancy that was declared but never built counts as absent:

- the resilience vital counts it as absent, so a complete over-claim scores 0 on D2 under the common-currency definition, and a partial shortfall scores the fraction actually delivered. See Section 4.4.
- the single-point-of-failure condition fires wherever the spec sheet promised none.

The record and any display must show the counterfactual: "declared Rx, built as Ry; as Ry it would score N", tagged DESIGN SHORTFALL, with re-declaration as the remedy. An honest, versioned new spec sheet starts a new score series. This penalizes only over-promising, because the spec sheet is the owner's own choice. A worked example is in Section 5.

A permanent shortfall can pin the headline at a low value for as long as it lasts. To keep day-to-day operational faults visible during that time:

- every binding reason (a reason that is currently holding the score down) is tagged DESIGN SHORTFALL, OPERATIONAL or PLANNED. See Section 4.8.
- a breadth indicator and the full set of vitals keep moving even while the headline cannot. See Section 4.5.

4.3 The Rulebook, fixed before measurement

Every score is computed against a named, versioned Rulebook. The first, RB-0.2, lists 40 parameters (T1–T40). Each Rulebook value, and each preset value, rests on one of four kinds of reference:

- **(a) Standard exists.** The IEEE 802.3 physical-layer budgets are the only case. The complete-PHY post-FEC frame loss ratio is 6.2×10^{-11} for 64-octet frames at 200G and above, and 6.2×10^{-10} at 100G, derived from the bit-error-ratio objectives of the IEEE 802.3 projects [31, 32, 33, 34, 35, 36]. The Ultra Ethernet specification reproduces them [37, 38]. They are used as distance to budget, not as alarm levels. On a spec sheet they are a floor: an owner may promise better, never worse.
- **(b) Best practice exists, graded by authority.** An example is the Ultra Ethernet method for estimating link reliability from FEC statistics [37]. Sources are graded by how widely they are recognised: a single vendor's or single operator's value is usable only as a declared reference for its own technology.
- **(c) Must be defined by Fabric Vitals.** The value is set only by named validation experiments, and it is frozen before it is used.
- **(d) Method only.** No universal number can exist. ECN marking, for example, depends on configured thresholds [4, 17]. The Rulebook then standardizes the procedure instead of the number: declared configuration, a measurement definition, a commissioning measurement under a standard test suite, the declared value and any preset acceptance criterion, and versioned re-registration on any change. Operators already work this way: one 800-GPU cluster fixed its ECN and PFC parameters after a vendor-validated sweep and left them unchanged [39].

Baselines become anomaly evidence. Rolling baselines and peer comparisons still run, but they only produce findings and trigger investigation. They never move a score. The reason is that a learned baseline turns whatever a fabric *is* into what it *should be*.

- One cloud validation system learns its pass criteria from the fleet's own results, with "no ground-truth on which nodes are defective" [1]. The inference that a uniformly degraded fleet would then set its own criteria is ours; that paper does not test it.
- Comparing one plane with its peers is valid only where the transport keeps the planes equally loaded [14].

In RB-0.2, 1 parameter has a standard behind it, 4 rest on graded best practice, 26 must be defined through experiments, and 9 are methods. Only one value is actually set: the IEEE budget. The Rulebook is a skeleton until validation fills it in.

4.4 The seven vitals

A common currency. Every vital is measured on one scale: the fraction of the declared promise being delivered, from 0 (none of it) to 100 (all of it), judged against the spec sheet. A resilience vital of 75 means three quarters of the declared resilience is available now; a congestion vital of 75 means three quarters of the promised loss and pause envelope is being kept. This is what makes the weakest-vital rule meaningful: the headline is the promise that is least kept. It is the design intent of the Rulebook's value functions and their anchors (parameters T25 and T31), whose shapes remain to be calibrated. See Section 4.5.

Vital	Question it answers	Main evidence
D1 Reachability	Can every endpoint that should reach every other get packets through?	Control-plane state compared with what was intended, including routing stability, <i>and</i> data-plane probes (the only way to see black holes)
D2 Resilience	If the next tolerated failure happens, will the fabric still do its job?	Remaining redundancy and residual capacity relative to the spec sheet; measured recovery and convergence time against the recovery objective
D3 Congestion and loss	Is traffic getting through without harmful loss, pausing or congestion spread?	Loss by cause, PFC pause time, pause storms and deadlock, buffer stress, ECN against its registered reference, NIC transport anomalies, lossless-configuration consistency
D4 Delay and tail	Is the network adding more delay than it should, at the tail?	Probe delay per path class against the tail-delay budget
D5 Physical link integrity	Are the links, optics and PHYs healthy?	Distance to the physical-layer budgets, link reliability from FEC statistics, errors, flaps, optics readings
D6 Effective capacity	Is there enough usable bandwidth where traffic goes, spread well?	Peak utilization in communication phases against declared capacity; imbalance on flow-hash fabrics
D7 AI communication	Tested the way AI jobs use it, does the fabric perform as promised?	Synthetic collectives and RDMA pair scans on known-good hosts

What is in scope. Only network-attributable evidence is scored: signals from switches, NIC ports and transport counters, running configuration, or active measurements that exercise the network independently of tenants' compute. Raw job signals are excluded because they mix in non-network causes:

- one collective-library error code covers GPU crashes, network disconnection, out-of-memory and user code [40];
- a host PCIe downgrade surfaced as a surge in PFC pauses [41].

Job signals enter only after attribution to a network element [7, 40].

Two named gaps, now explicit.

- **Lossless-transport configuration consistency.** Lossless RDMA needs PFC priorities, ECN marking, the DSCP-to-traffic-class mapping and MTU to agree across switches and NICs. Misconfiguration is a documented failure cause: in Meta's lossless fabrics, congestion drops were "uncommon and observed mostly due to misconfiguration" [4], and Microsoft traced one production incident to "the buffer misconfiguration of a newly introduced switch type" and runs a service that checks running against desired configuration on switches and

servers [16]. The declared configuration is a spec-sheet field and a conformance check; a live mismatch is D3 evidence. It is not a new critical condition, because its harmful consequences are already capped by the loss, storm and black-hole conditions.

- **Routing stability and convergence time.** Session flaps and route churn are named as D1 control-plane evidence. Routing convergence time is part of the recovery objective and is judged by D2, measured in controlled-failure tests and, where probes allow, on real events. The IETF AI-fabric benchmarking draft already lists FIB and route convergence time, the time to converge routing after a topology change, among its fabric health indicators [42].

The D7 exception. D7 only corroborates the other vitals unless the class requires it; of the presets, only R3 does. Its cost is bounded by method: short runs, a small reserved set of nodes or scheduled gaps between jobs, and a frequency bound, with the values not yet set. This avoids a perverse incentive: if D7 simply joined the minimum whenever it was run, a fabric that tests itself could only ever score the same or lower than one that does not.

Scope and deliberate exclusions. v0.x scores the back-end scale-out fabric, which may use the TRAINING, INFERENCE or GENERAL workload profile. Security is excluded. Separate front-end networks, inference-serving network domains, storage/checkpoint networks, scale-up interconnects (the links that join GPUs inside one server or rack) and multi-tenant isolation are out of scope in v0.x and are candidate future profiles. See Section 8.

Vendor-neutral telemetry is uneven.

- Physical-layer counters are well covered by the IEEE and OpenConfig data models [43, 44].
- PFC counters and buffer watermarks exist in the SAI switch interface but not in OpenConfig, and the PFC counters in IEEE 802.3.2 are deprecated [43, 44, 45]. PFC configuration has no OpenConfig model either [46].
- RDMA NIC counters have no vendor-neutral set; their names and meanings differ from one driver to another [47].

The congestion vital therefore depends on work outside this proposal. The dependency is stated, not hidden.

4.5 The weakest-vital headline and critical conditions

The Vitals Score is the minimum across the vitals the spec sheet requires. No weights are used. Precedents:

- the IETF Quality of Outcome draft scores by the minimum over its sub-scores [48];
- ITU-T M.3042 takes a minimum across types of network element inside 8 of its 21 third-level indexes [49];
- US bridge-condition rules rate each bridge by its lowest component rating [50];
- facility tiers rate a site by its lowest subsystem [30].

The minimum is not assumption-free. It assumes the vitals are commensurable, meaning fairly comparable with each other: a 60 in physical integrity must be as bad as a 60 in congestion. The common currency makes equal values mean the same share of each promise delivered *by definition*. Whether that also means equal operational impact is an open empirical question, tested by experiment. The minimum also ignores non-binding vitals, however close they are. So every headline is printed with:

- a **breadth indicator** ("k of 7 vitals not delivering their full promise");
- the **next-binding vital**, the one that would become the headline if the lowest were fixed;
- trends on the full set of vitals.

Critical conditions cap the score whatever the vitals show. A cap holds the score at or below a fixed low value while the condition lasts; each condition names itself as the reason:

Condition	Why it matters
CC1 Partition or black hole	Black holes and silent drops are invisible to device state [5, 14]

Condition	Why it matters
CC2 PFC storm or deadlock	"a single malfunctioning NIC may block the entire network" [16]
CC3 Sustained loss on a lossless class	"normally no RDMA packets should be dropped" [16]
CC4 Single point of failure on job-bearing capacity	Applies only where the spec sheet promises none [13]
CC5 Redundancy exhausted	In one testbed, all-reduce slowed down once more than half of a switch's <i>redundant</i> uplinks were down [1]
CC6 Uncorrectable errors beyond the IEEE budget	The physical-layer budget above, measured as a link reliability ratio [37]

A condition caps the score only when the share of job-bearing capacity it affects reaches a **blast-radius threshold**, parameter T16. The cap value (parameter T15), the persistence time and the blast-radius threshold are Rulebook parameters, all "value not set".

4.6 Confirmed score and range

Unmeasured capacity is treated as unknown. Each vital is computed twice over the *declared* inventory, not just the part that is reporting:

- once with every unmeasured device or time slice counted as 0, giving the **lower bound**;
- once with every unmeasured device or time slice counted as 100, giving the **upper bound**.

The Vitals Score is the confirmed lower bound, printed with the upper bound and the coverage (the share of the fabric that was measured). Critical conditions count unmeasured capacity as affected. Once raised, a condition clears only when it is observed to have cleared, never because telemetry stopped arriving.

Removing telemetry cannot raise the score. Removing telemetry can only lower each lower bound and can only increase the affected share for each condition, and the minimum is monotone. A simpler rule that excludes unmeasured areas fails this test. In an illustrative case, hiding three bad switch groups *raises* the score from 69 to 80. See Section 5.

Two refinements.

1. **Grace period.** A restart of the collector (the system that gathers telemetry) or a missed poll is not a fabric event. A sample therefore becomes "unmeasured" only after it is older than its declared sampling interval plus a **grace period G**, parameter T30. During the grace period, counters use their last completed interval and gauges carry their last value forward. The price is bounded: removing telemetry can never raise the score above what full telemetry would have shown at some moment within the preceding G. G = 0 restores the strict property. The value of G per metric type is to be set by an experiment that measures false drops caused purely by telemetry gaps.
2. **Coupling the coverage floor and the blast radius.** Below a **coverage floor F**, parameter T29, the result is UNSCORABLE. Could a fabric that passes the floor still be capped "critical" purely because some capacity is unmeasured? Unmeasured share u satisfies $u \leq 1 - F$, and it alone reaches the blast-radius threshold only if $u \geq T16$. Both hold together exactly when $F + T16 \leq 1$. So, by default, every critical condition must satisfy $F + T16 > 1$, with both shares measured on the same capacity base (constraint T39). A Rulebook may relax this for a specific condition only with a written rationale.

The consequence is intended. Hiding a region large enough to trigger a condition necessarily pushes coverage below the floor, so the fabric becomes UNSCORABLE instead of escaping the cap. Fabrics with thin telemetry, including many small clusters, will be UNSCORABLE for affected conditions until their telemetry improves.

Evidence-quality confidence is a separate figure that says how far the evidence can be trusted. It covers sample age within the grace period, resolution, platform capability, counter integrity [51] and whether counters mean the

same thing across vendors [47]. It never changes the number.

4.7 UNSCORABLE, with a missing-telemetry list

When coverage falls below the floor, the result is **UNSCORABLE**, meaning "conformance not confirmed". It ranks below every numeric score and is never shown as a pass. This follows the IETF service-assurance convention that -1 means "no value could be computed" [52], and the credit-scoring practice of declaring thin files unscorable [53].

An UNSCORABLE result must list exactly which required telemetry is missing, and where: vital or condition, device, metric, capacity share, and whether each item is missing, stale beyond the grace period, or unsupported by the platform. The result is an instruction to the operator, not a dead end.

4.8 Availability and score bands

Availability. Owners and customers often ask a time question: "how much of the time was the fabric healthy?" Fabric Vitals answers it with one definition. Over a period divided into scoring windows, availability is the share of windows in which the Vitals Score is in the healthy band and no critical condition is active. "Healthy X% of the time" means exactly that. UNSCORABLE windows count as not healthy, because unknown is not healthy, and their share is reported beside the figure. The owner declares an availability target on the spec sheet; availability is reported against it as met or not met. Availability accompanies the score and never changes it. No target values are proposed.

Reading the score over time. A Vitals Score is a reading, not a verdict. It is computed once per scoring window, and each reading stands on its own: it says what the fabric confirmed in that window and why. Reading it well means reading it three ways. Now: the current score and its binding reason, for whoever is on call. Over a period: availability as defined above, for whoever must report whether the fabric kept its promises this week or this month. Direction: the series of readings, for whoever must see slow decay before it becomes an incident. A display of the series shows, on the same axis, the upper bound as a band above the confirmed score, so that a widening band shows lost telemetry rather than lost health; each critical condition as an interval from raised to cleared; UNSCORABLE as a gap, never as zero; and every change of spec sheet or class, so that a jump caused by a changed promise is not mistaken for a change in the fabric. The series is never smoothed or averaged, for the same reason the vitals are not averaged: a two-hour black hole must not vanish into a 30-day mean. Every marked change is derived from the score's own record: the binding reason that changed, the condition that was raised or cleared, the coverage that crossed the floor, the spec sheet that was re-declared. An operator may attach a note to a marker; a note never replaces one and never creates one. Readings, with their binding reason and coverage, are kept for at least the longest reporting period offered.

Score bands. Bands are a display rule, not part of the score. The Rulebook parameter T40 holds two band edges (value not set):

- **green** at or above the upper edge;
- **amber** between the edges;
- **red** below the lower edge;
- **red**, regardless of the number, while any critical condition is active;
- **grey** for UNSCORABLE.

The cap value lies below the lower edge, so a capped score is red by number as well as by rule. Bands are part of the Rulebook and are published only once calibrated against measured impact. Until then, any displayed bands are illustrative and must be labelled so; demonstrations of the method use illustrative edges. Voice-quality bands look similar but belong to a different scale and must not be borrowed [54].

Planned maintenance. Maintenance reduces the resilience a fabric actually has while it lasts, so the score reflects it. When a maintenance window was declared in advance (declarations are versioned, like the spec sheet), the

affected contributors are tagged **PLANNED**. The spec sheet states whether declared windows count against the availability target. There is no undeclared "maintenance mode": nothing suspends scoring.

4.9 Every score explains itself

- **The top-level explanation is exact.** Because the score is a minimum, it always equals exactly one named vital or one named critical condition, so the top reason is never a blend.
- **Contributors follow.** Below the top level, the contributing faults are listed in order, following the credit-score practice of listing the factors with "the most influence first" [55]. This also satisfies the IETF rule that "a non-maximal score must always be explained by one or more symptoms" [52].
- **Full lineage.** Every value can be traced back to the raw counters, the value-function version, the Rulebook entry and the spec-sheet field it was judged against.
- **Presentation.** A score is never shown without its class. The unit of presentation is "Vitals Score N, class Rx (spec sheet version)"; a bare number is non-conformant. Scores are whole numbers, shown with their band.
- **Language models** may turn the structured result into prose. They never compute or change it.

Scores are comparable for the same fabric over time (under the same spec-sheet version), and between fabrics on the same class preset, or on identical spec sheets, under the same Rulebook version. A score for one class is not a ranking against another: it only confirms conformance to its own spec sheet.

4.10 Scoring scope and large fabrics

A job runs inside a pod or scalable unit, and it is gated by the slowest transfer there [13, 14]. A Vitals Score therefore applies to a declared **scoring scope**: that unit, stated on the spec sheet. A fabric with many pods is reported as a **set of scope scores**, one per pod, with their distribution and the lowest scope named, never as one number blended by capacity; a blended figure would let one failing pod disappear into a large healthy fabric, which is the averaging problem again. Coverage floors and blast-radius thresholds are evaluated within the scope.

4.11 Starting on day one

An operator can start with the **Day-One Set**: the smallest set of vitals, counters and probes that yields a valid Vitals Score, drawn from telemetry most platforms already export. It covers every vital with standard counters. For reachability, resilience and delay: interface state, routing sessions, cabling checks and a probe mesh. For congestion: drops, switch-side ECN marks and, on lossless fabrics, PFC pause time. For the physical layer: FEC, CRC and flap counters. For capacity: interface octets. A score computed from it is a Vitals Score like any other, under the same Rulebook. Richer telemetry (NIC transport counters, buffer watermarks, in-band telemetry and optics) is optional: where it is not observed, it is listed as unscorable at the metric level and reflected in evidence-quality confidence, never in the score.

5. A worked example

Every number in this section is ILLUSTRATIVE.

The masking example. Four vitals are at 100 and one is at 10. The arithmetic mean is 82. So is the result of any scheme that adds up across areas, including M.3042's two-stage form, which takes minimums inside areas but sums across them. The Vitals Score is 10, with the binding reason named.

A fabric under stress.

- The vitals are D1 100, D2 90, D4 85, D5 97, D6 80.

- For congestion (D3), the fabric has ten leaf groups, each carrying 10% of the relevant capacity. Seven are at 95. Three are under PFC stress at 10 and are in an active pause storm (CC2).
- Using a simple capacity-weighted mean for illustration, $D3 = 69.5$.
- The storm affects 30% of capacity. Against an illustrative blast-radius threshold of 25%, the condition fires, so the score is capped and names CC2.

The operator hides the three stressed groups.

Rule	What happens
Excluding unmeasured areas (the rejected design)	The groups vanish, D3 becomes 95, and the headline <i>rises</i> to 80
Fabric Vitals, coverage floor 80%	D3 coverage falls to 70%, below the floor: the result is UNSCORABLE, listing the three leaf groups and their missing congestion counters
Fabric Vitals, coverage floor 65% (here the illustrative $F + T16 = 0.90$, which breaks the coupling rule)	The confirmed D3 is 66.5 and the unmeasured 30% still triggers CC2, so the operator gains nothing. But with this setting, three healthy groups that were merely <i>unmonitored</i> would also cap the score, which is why the default Rulebook forbids it

A collector restart. All congestion telemetry arrives 40 seconds late, with a 10-second sampling interval.

- With no grace period, coverage collapses and the fabric falsely becomes UNSCORABLE.
- With a 60-second grace period, nothing changes.

Over-claiming. An owner declares class R2 (dual-homed) for a 256-NIC scoring scope.

- *Complete over-claim:* every NIC is in fact single-homed, so the fabric is built as R1. None of the declared second attachments exists, so D2 delivers none of its declared promise: $D2 = 0$, and the Vitals Score is 0, class R2, tagged DESIGN SHORTFALL. The record adds: "declared R2, built as R1; as R1 it would score 91". The remedy is to build the second attachments or re-declare as R1.
- *Partial shortfall:* 64 of the 256 NICs are single-homed. Three quarters of the declared dual-homing exists, so the attachment part of D2 is 75. Because the 25% of single-homed capacity reaches the illustrative blast-radius threshold, the single-point-of-failure condition also fires, which is the state in the record below.

The explanation record for that capped state, with illustrative band edges of 80 and 60:

Vitals Score 41, class R2 (spec sheet v2) · confirmed; ≤ 44 if unmeasured capacity is healthy · RED (illustrative band edges 80 and 60)
 Scope: pod-3 · fabric scope scores 41 / 83 / 88 / 90 (lowest: pod-3) · Rulebook RB-0.3 (hypothetical) · 15-minute window
 Class: declared R2 · verified R1 · NON-CONFORMANT (64 NICs single-homed; spec sheet declares dual-homing)
 Counterfactual: declared R2, built as R1; as R1 it would score 58 [DESIGN SHORTFALL]
 Breadth: 3 of 7 not delivering their full promise · next-binding $D3 = 58$ · evidence-quality confidence 90%
 Binding: CC4 single point of failure [DESIGN SHORTFALL]
 1. D3 PFC pause-time on leaf-12/13 spine ports exceeds the pause budget [OPERATIONAL]
 2. D3 NIC out-of-sequence rate behind leaf-12 exceeds the loss budget [OPERATIONAL]
 3. D2 spine-2 drained in declared maintenance window MW-7 [PLANNED]
 Anomaly evidence: ECN ratio 4× its registered commissioning value
 Availability, last 30 days: healthy in 96% of windows (0.5% UNSCORABLE) · declared target 99% (declared maintenance counts: none)
 Remedy: restore dual-homing, or re-declare as R1 (versioned; new score series)

6. Why not a metric checklist?

A common way to describe AI-fabric health is a checklist: NIC speed, bisection bandwidth, oversubscription, tail latency, loss, ECN and PFC activity, utilization, job completion time and, for inference, tokens per second. The IETF benchmarking drafts express this view for the lab: the back-end fabric "determines Job Completion Time (JCT), training throughput, and accelerator utilization" [42], and for inference serving it determines time to first token,

inter-token latency "and aggregate throughput in tokens per second (TPS)" [56]. Fabric Vitals uses many of these metrics as inputs. A checklist on its own, however, falls short in five ways.

- **It gives no verdict.** A list of numbers does not say whether the fabric is healthy; the benchmarking drafts explicitly leave "acceptance criteria or performance requirements" out of scope [42]. Fabric Vitals gives one verdict and explains it.
- **It mixes build with operation.** NIC speed, bisection and oversubscription describe the build; loss, pauses and delay describe operation. Fabric Vitals puts the build on the spec sheet, checks it once by conformance, and scores operation continuously.
- **It judges design choices universally.** A checklist implies that more bisection bandwidth or less oversubscription is always healthier. Rail-only fabrics are deliberately designed without any-to-any paths [20]. Fabric Vitals judges each fabric against its own declared promises.
- **It counts workload performance.** Job completion time and tokens per second depend on accelerators, software, models and schedulers as well as the network, and job-level signals mix causes: one collective-library error code covers GPU crashes, network disconnection, out-of-memory and user code [40]. Fabric Vitals scores only network-attributable evidence, and uses job metrics for validation.
- **It has no rule for missing data.** Unmonitored links silently drop out of a checklist. In Fabric Vitals they count as unknown, and below the coverage floor the result is UNSCORABLE, with a list of what is missing.

7. Relationship to existing work

ITU-T M.3042 (2025) is one principal prior work. It is the ITU-T framework for communication network health, and it defines a network health index as "a single summary indicator expressed in quantitative terms indicating the degree of communication network health" [49].

- **Structure.** Four levels: an overall index; four dimensions (availability, reliability, security and maintainability); 21 third-level indexes; and scenario-specific indexes.
- **Aggregation.** The top two levels are weighted sums, with weights "outside the scope of this Recommendation". Inside 8 of the 21 third-level indexes, M.3042 already applies the weakest-element rule: it takes the minimum across network-element types.
- **Where Fabric Vitals differs.** It applies that minimum across its dimensions, adds critical conditions, scores against a declared spec sheet and a pre-registered Rulebook rather than operator-chosen weights, and uses AI-fabric inputs. It also excludes security, which M.3042 includes, to stay focused on fabric health.

The two are complementary. Fabric Vitals could be described to ITU-T readers level by level against M.3042.

IETF AI-fabric benchmarking drafts [42, 56, 57]. These are individual drafts (proposals by their authors, not yet adopted by an IETF working group) that define *Fabric Health Indicators* for lab benchmarking, and they explicitly leave acceptance criteria out of scope. Their indicators map naturally onto Fabric Vitals inputs: PFC events and storms, ECN ratio, loss, buffer occupancy, retransmission, link flaps, CRC and FEC errors, and route convergence time. Their benchmark methodology is a candidate basis for the commissioning test suite.

The IETF Link Quality Score draft [58] is the closest published score for AI fabrics that the author found. It defines a vendor-agnostic, per-link, 0–255 composite built from utilization, queue depth, ECN, PFC pause and drops. It feeds adaptive routing, and "the computation of LQS from hardware signals is EXPLICITLY implementation-defined".

- **What it is not.** It scores individual links for congestion; it is not a fabric-wide health measure.
- **How Fabric Vitals can use it.** It is a possible input to the congestion and capacity vitals. Its monotonicity requirement (a worse input must never give a better score) matches ours, and its caution that two conforming implementations may disagree describes exactly the problem a pre-registered Rulebook addresses. Its "healthy"

range is not adopted as a reference.

IETF service assurance (SAIN) and Quality of Outcome. SAIN, the IETF's service assurance for intent-based networking architecture, standardizes a 0–100 health score with –1 for "no value could be computed", and requires symptoms for any non-maximal score [52, 59]. Fabric Vitals adopts compatible conventions for UNSCORABLE and explanation; those ideas are prior art, and its vitals could be carried as SAIN subservices. The IETF Quality of Outcome draft scores by the minimum with application profiles [48]. Its requirement-relative normalization and minimum aggregation are also direct prior art for the v0.2 proposal, whose cross-vital commensurability remains to be validated.

Facility tier practice.

- **Uptime Institute's Tier Standard: Topology** explicitly excludes IT and the network. It rates a site by its lowest subsystem, with no partial tiers, confirmed by outcome-based tests [30].
- **ANSI/BICSI 002** (a data-center design standard) defines network infrastructure classes that the owner declares before design [29].
- **ANSI/TIA-942** and **EN 50600** define classes for cabling infrastructure [26, 27].

Fabric Vitals borrows the ideas of declaration, weakest-subsystem rating and outcome testing. It does not claim to implement those standards' requirement texts, which the author has not consulted.

8. Limitations and what is not yet known

No values yet. Of the 40 parameters, one has a standard behind it (the IEEE budget). Twenty-six must be defined by experiment, and nine are methods. The class presets carry no promise values yet. Until validation runs, a Vitals Score is for discussion and experimentation, not for acceptance decisions.

The evidence is concentrated.

- The mechanisms of failure are well evidenced.
- "Normal" ranges and failure rates are evidenced almost only at hyperscale. Most of the empirical evidence comes from a few hyperscale operators, and one heavily cited source is a preprint whose results depend on its multipath transport [14].
- At 32 to 2,000 GPUs, where many readers operate, public evidence is thin. The two operational reports found cover 504 and 800 GPUs, and neither collected fabric congestion counters [39, 60]. Vendor validation results at those scales are private [24, 61].
- Every value carried over from hyperscale is marked "not validated at this scale".

Telemetry gaps.

- The congestion vital depends on vendor-neutral PFC, buffer and NIC counter definitions that do not yet exist.
- The lossless-configuration check can read the switch-side MTU, DSCP classification and whether ECN is enabled through vendor-neutral models, but not the PFC configuration or NIC-side settings.
- Because the coverage rule is strict, many small fabrics will be UNSCORABLE for affected conditions at first. They will receive a list of what to add.

Standards texts. The requirement texts of several facility and cabling standards, and the normative clause text of IEEE 802.3, were not consulted. The physical-layer budget is cited from the IEEE project objectives and the Ultra Ethernet specification.

Design risks to test.

- The minimum's commensurability assumption may fail.
- The grace period trades false drops for bounded staleness.
- Scoring against the declared spec sheet can pin the headline during a design shortfall.
- Custom spec sheets can promise little; their scores are then honest but not comparable, which is why presets matter.
- Method-only references can lock in the behaviour of a fabric that was already bad at commissioning, wherever no declared value or acceptance criterion exists.

Scope and roadmap. v0.x scores the back-end scale-out fabric. That back-end fabric may use the TRAINING, INFERENCE or GENERAL workload profile. Separate front-end networks, inference-serving network domains, storage/checkpoint networks, scale-up interconnects (NVLink-class, including Ethernet-based versions) and multi-tenant isolation are out of scope and are candidate future profiles. Security is excluded.

The field is moving. Relevant prior art predates Fabric Vitals, including IETF SAIN, ITU-T M.3042, IETF Quality of Outcome, vendor assurance/health systems and facility-rating practice; additional AI-fabric scoring and benchmarking work continues to appear. Fabric Vitals claims potential novelty only in the proposed combination, not in those individual ideas. Related work must be re-checked before any submission.

How validation will proceed.

1. **Expert critique** of the class presets and the spec-sheet template, in public.
2. **Without GPUs, in a container-based emulated fabric.** Tests of the scoring machinery run on replayed or emulated telemetry: anti-gaming under telemetry removal, false drops from telemetry gaps, determinism, monotonicity, and detection of configuration mismatches and convergence time. They set the first Rulebook values: the coverage floor, the grace period, the coupling-rule values and the event definitions. An emulated fabric cannot reproduce switch-chip buffers, PFC or ECN at line rate, FEC and optics, or GPU collectives, so those values wait.
3. **With GPUs and hardware.** Lab experiments at 32 to 256 GPUs follow, including a test of whether equal vital values mean equal impact, the calibration of band edges, and the preset promise values. Shadow deployments (running the method alongside a live fabric without acting on its results) on mid-sized production fabrics come after that.

9. How to take part

The method, the Rulebook skeleton, the spec-sheet template and class presets, the metric definitions and the validation programme are published as an open specification at <https://github.com/fabric-vitals/spec>. The questions we most want answered are listed there as seed issues (starting points for discussion on the project's issue tracker): the two named gaps, the commensurability assumption, the role of D7, the class-preset proposal, evidence from mid-sized clusters, and band calibration. Three kinds of contribution are invited through that issue tracker:

- **Method critique:** the Rulebook / spec-sheet split, the class presets, the aggregation rule, the coverage rules.
- **Telemetry availability:** which counters and configuration your platforms export by default, and at what resolution.
- **Validation partnership:** lab or production fabrics, especially mid-sized clusters, that can help set the Rulebook and preset values.

Rulebook values will be fixed before measurement, through an open, versioned change process, and never tuned after the fact. "Fabric Vitals" and "Vitals Score" are claimed as marks of datacenternetwork.ai so that the names always mean the published method; anyone may state that a product implements Fabric Vitals when it follows the

published Rulebook and spec-sheet rules, and the marks will move with stewardship if it passes to a neutral body.

10. Disclosure

The author organization develops network software. Fabric Vitals is currently author-stewarded by datacenternetwork.ai and open to contribution; it does not yet have neutral institutional governance. The proposal is designed to be implementable by vendors and operators across platforms, but complete cross-vendor telemetry equivalence and score comparability remain part of the validation programme. The method, its Rulebook and its proposed class presets are offered for open discussion.

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